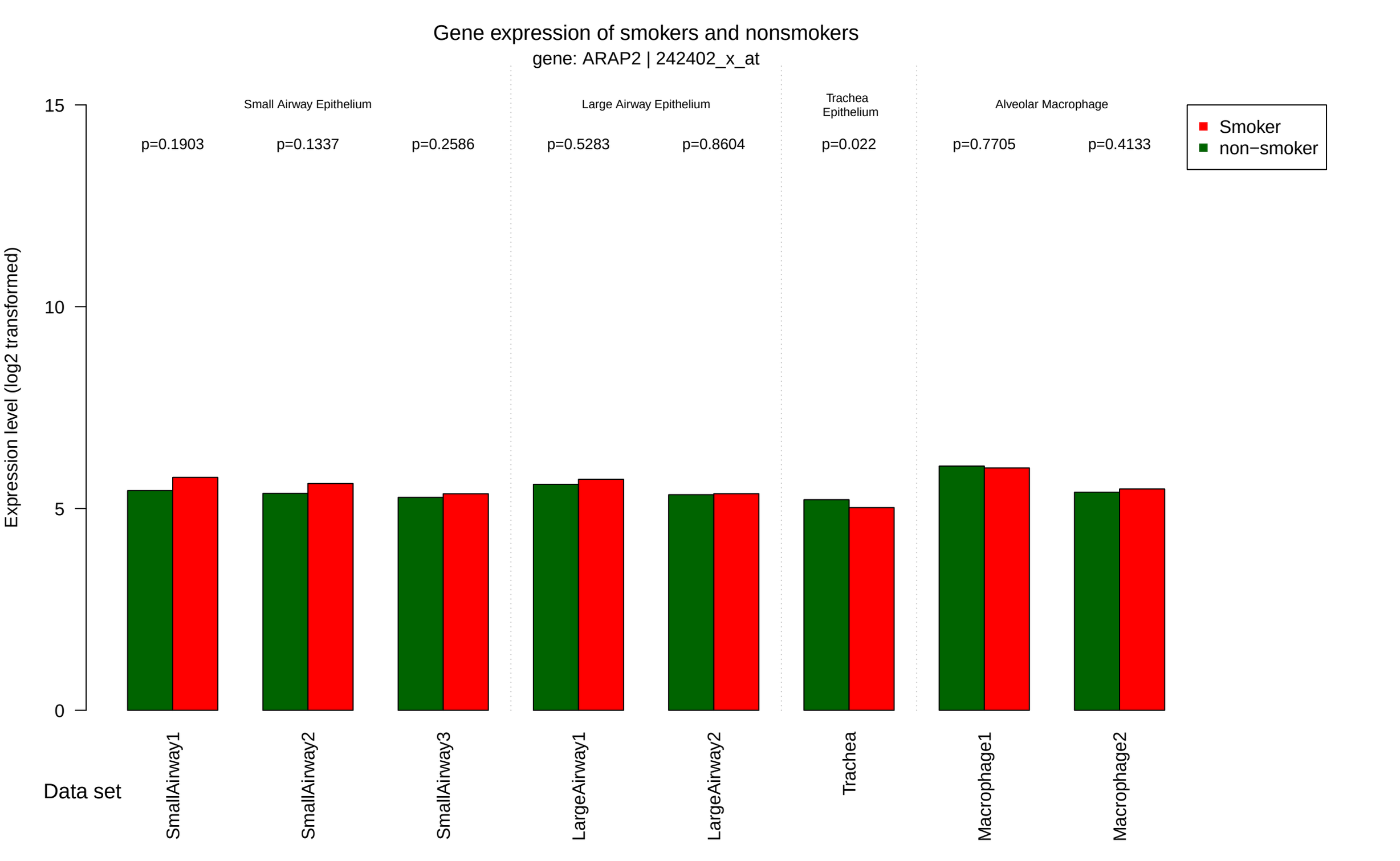
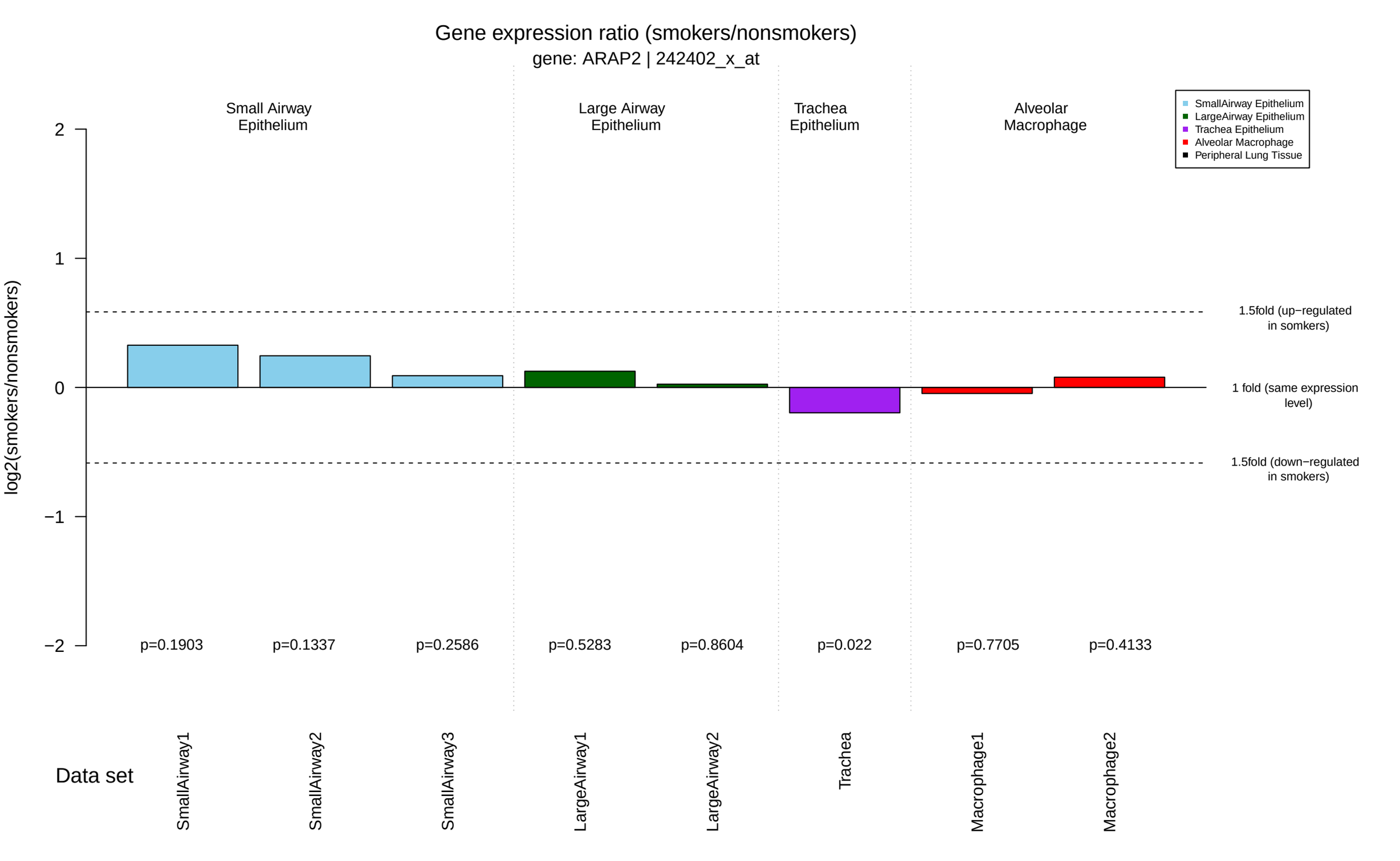
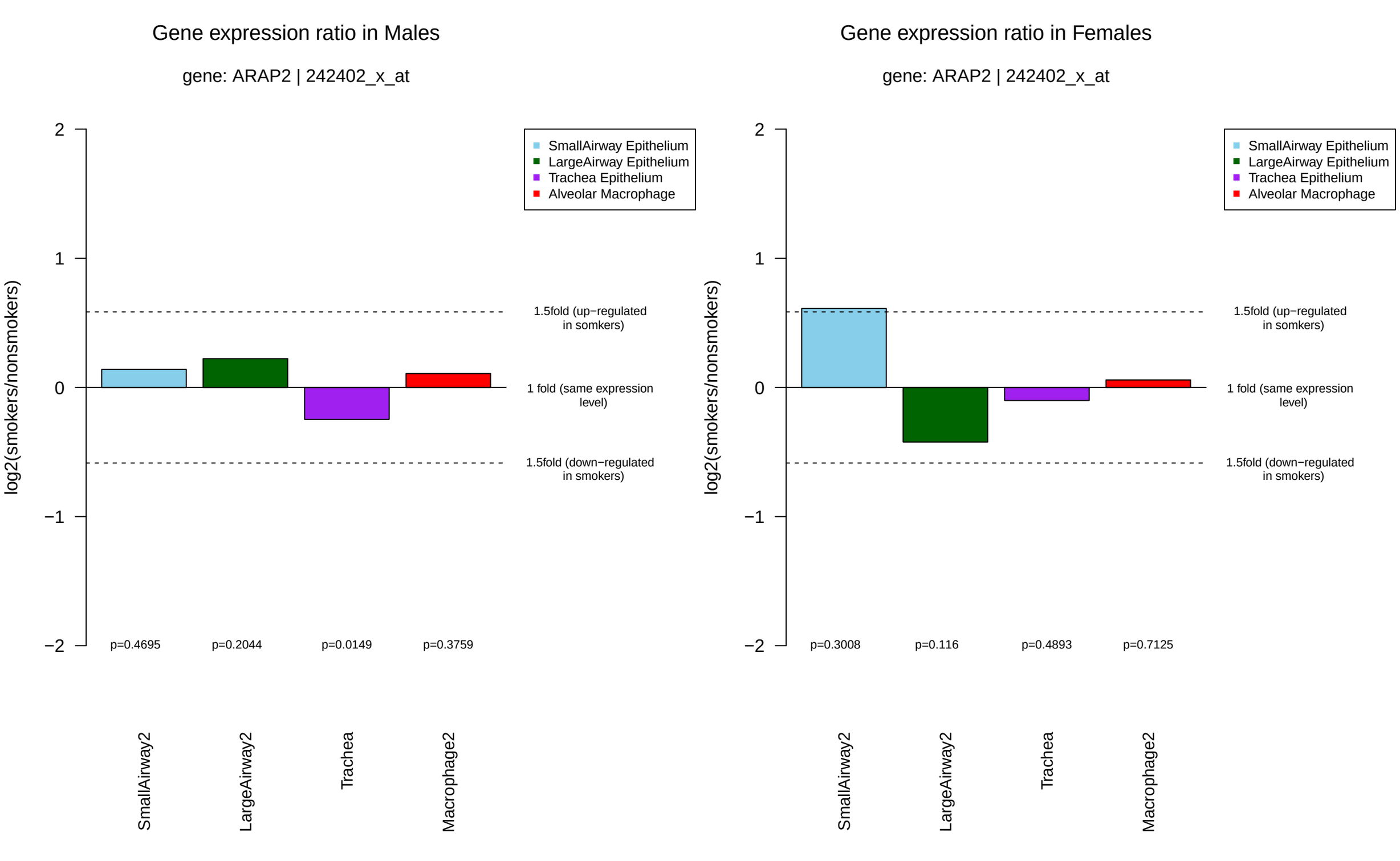
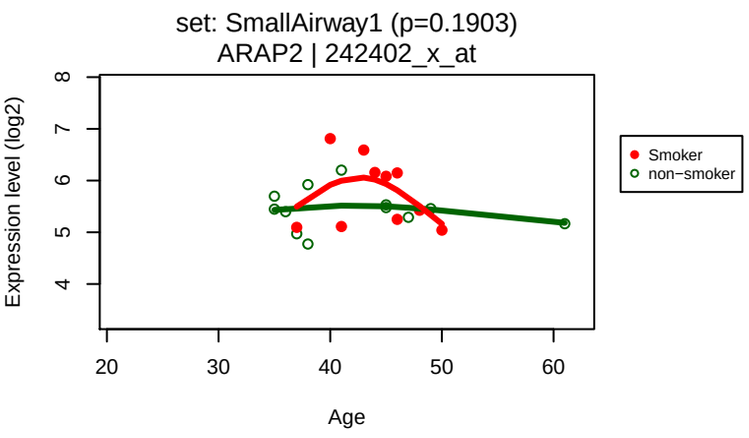
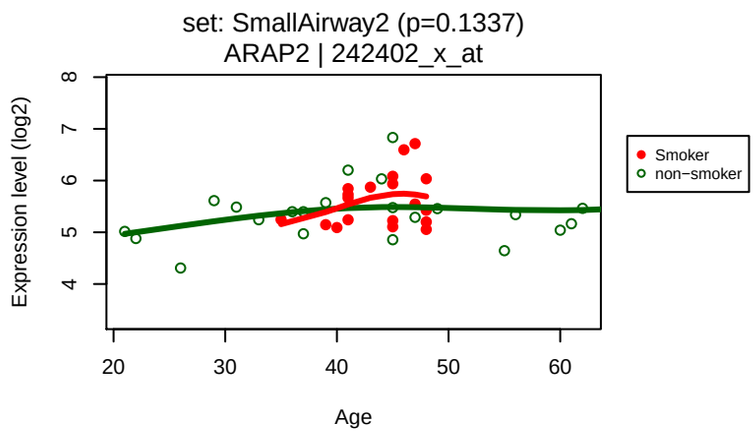
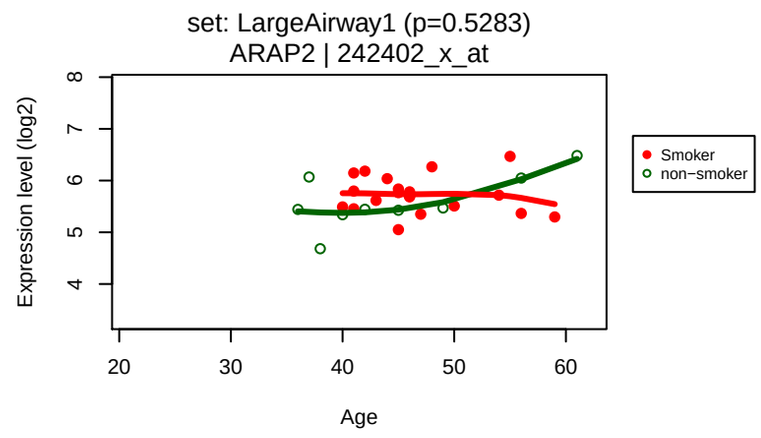

Gene expression of smokers and nonsmokers
gene: ARAP2 | 242402_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1903
p=0.1337
p=0.2586
p=0.5283
p=0.8604
p=0.022
p=0.7705
p=0.4133
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

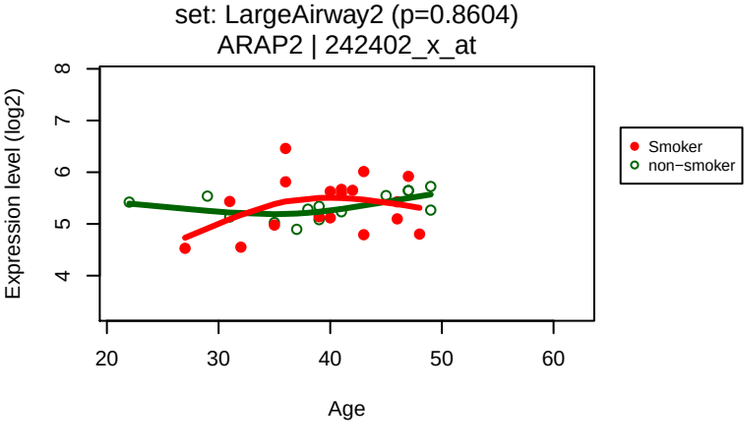

Gene expression ratio (smokers/nonsmokers)
gene: ARAP2 | 242402_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1903
p=0.1337
p=0.2586
p=0.5283
p=0.8604
p=0.022
p=0.7705
p=0.4133
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

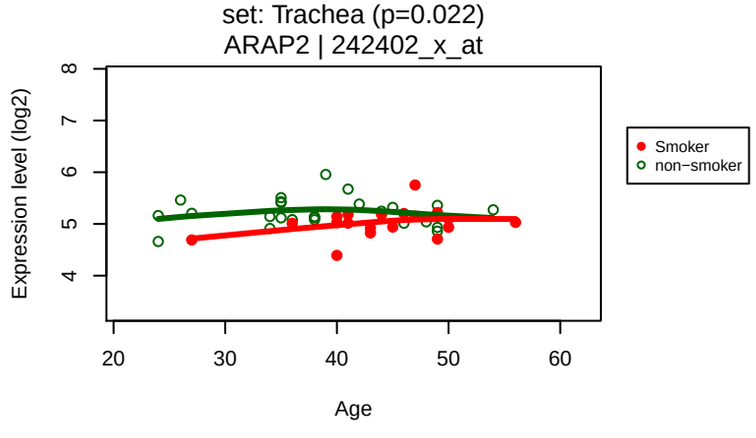

Gene expression ratio in Males
gene: ARAP2 | 242402_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4695
p=0.2044
p=0.0149
p=0.3759
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP2 | 242402_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3008
p=0.116
p=0.4893
p=0.7125
SmallAirway2
LargeAirway2
Trachea
Macrophage2

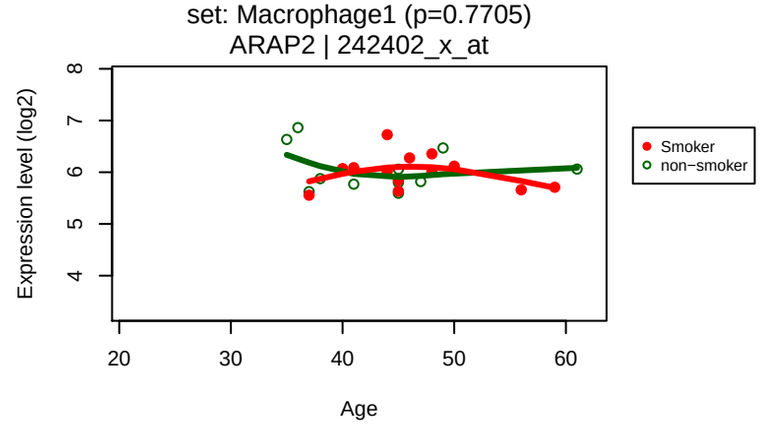

set: SmallAirway1 (p=0.1903)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker

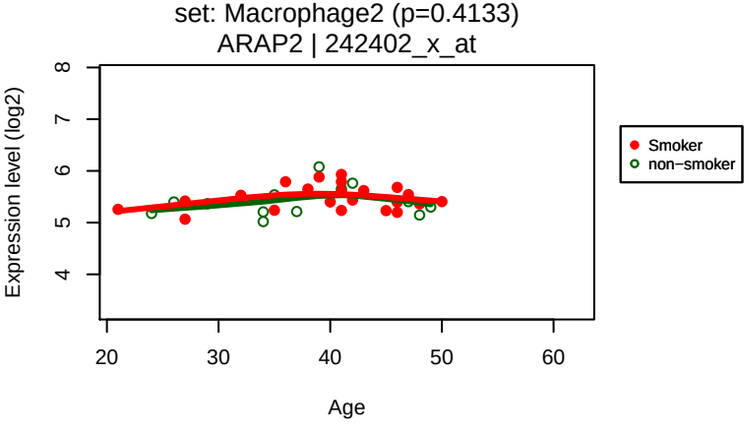

set: SmallAirway2 (p=0.1337)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker

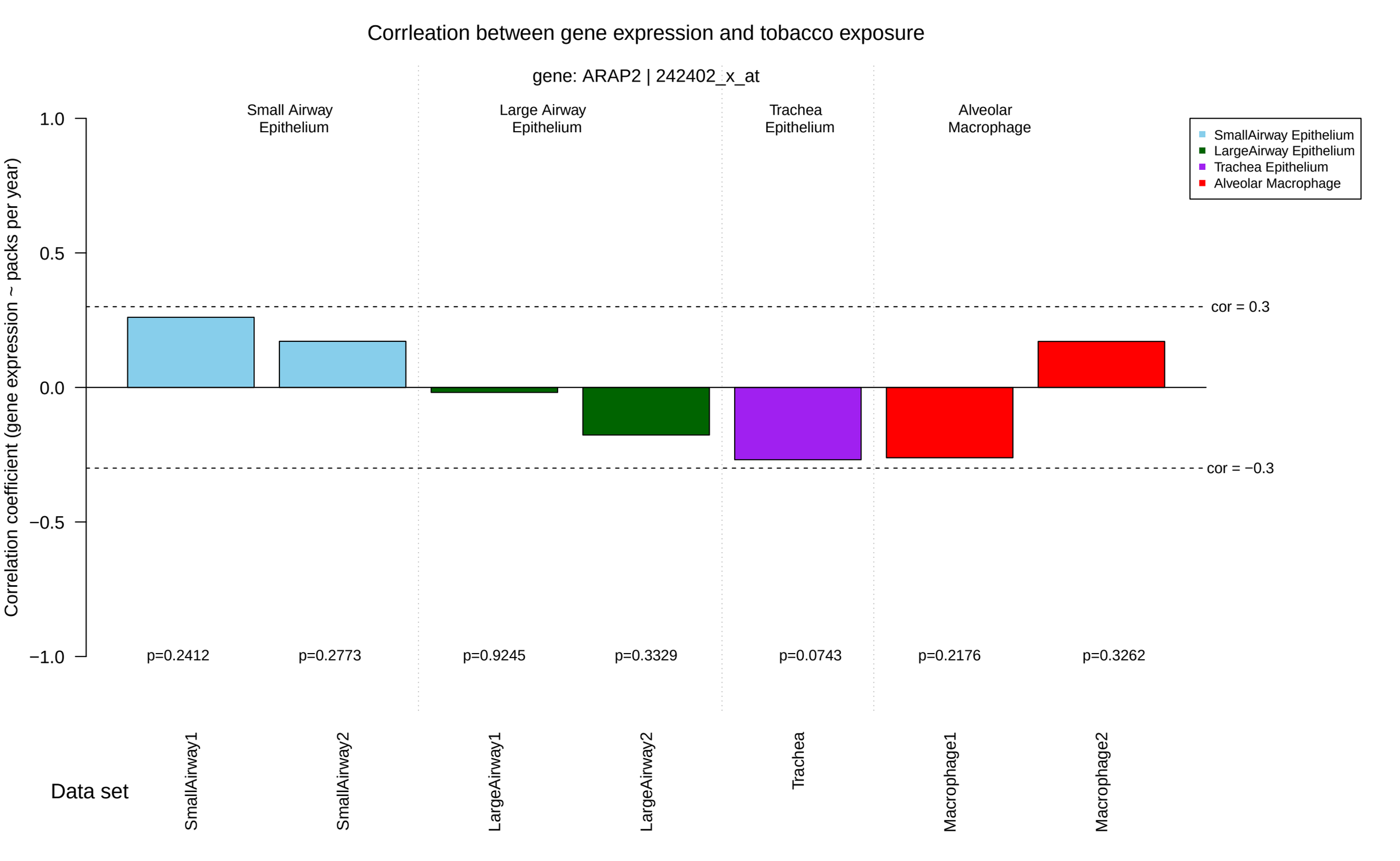

set: LargeAirway1 (p=0.5283)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8604)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.022)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7705)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4133)
ARAP2 | 242402_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP2 | 242402_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2412
p=0.2773
p=0.9245
p=0.3329
p=0.0743
p=0.2176
p=0.3262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2